Experiment 2

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data into a pandas DataFrame
file_path = 'C:\\sales_data.csv'
df = pd.read_csv("sales_data.csv")

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill missing Sales values with the mean
df['Sales'].fillna(df['Sales'].mean(), inplace=True)

# Drop rows with missing Product, Quantity, or Region
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by Product and calculate total Sales and Quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')

# Line plot of sales over time
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Sales by Region and Product
pivot_table = pd.pivot_table(df, values='Sales', index='Region',
                             columns='Product', aggfunc='sum', fill_value=0)
```

```
print(pivot_table)

# Correlation between numeric columns
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

# Heatmap of the correlation matrix
plt.figure(figsize=(8, 5))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

✓ 2.7s

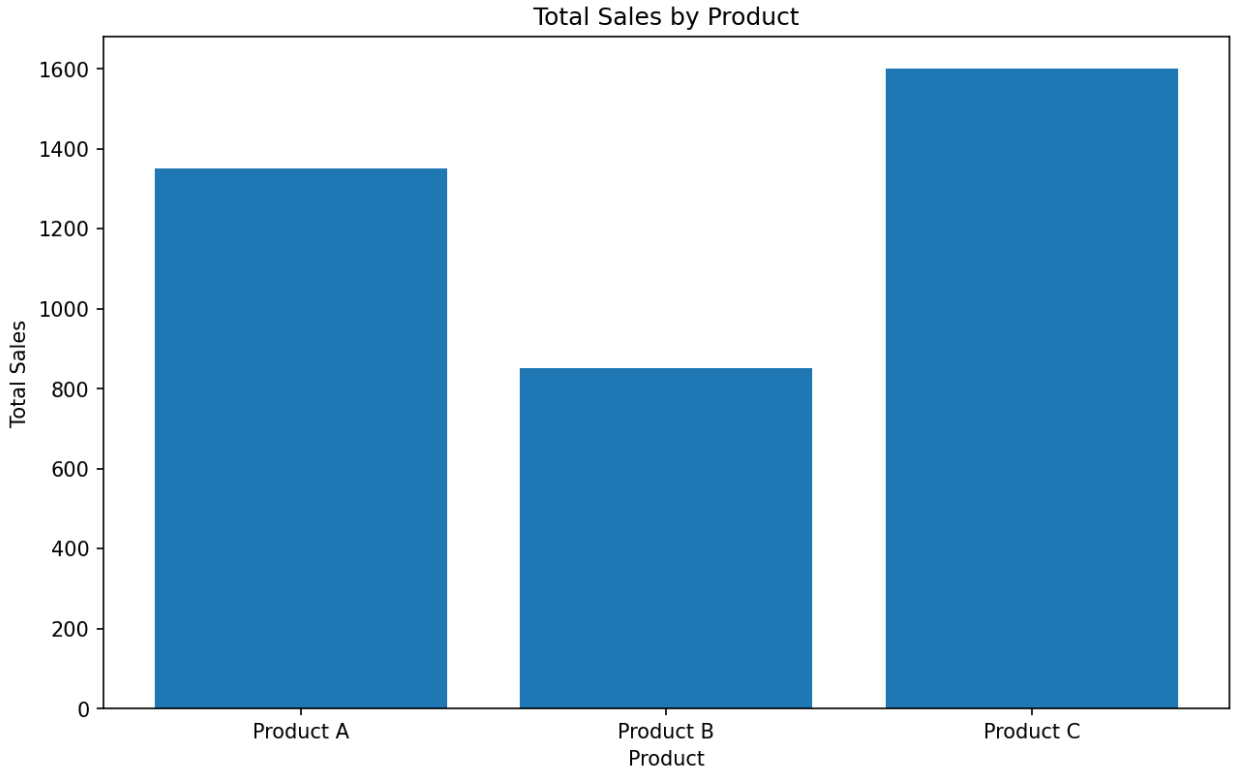
```
         Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6   East
4  05-01-2023  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
```

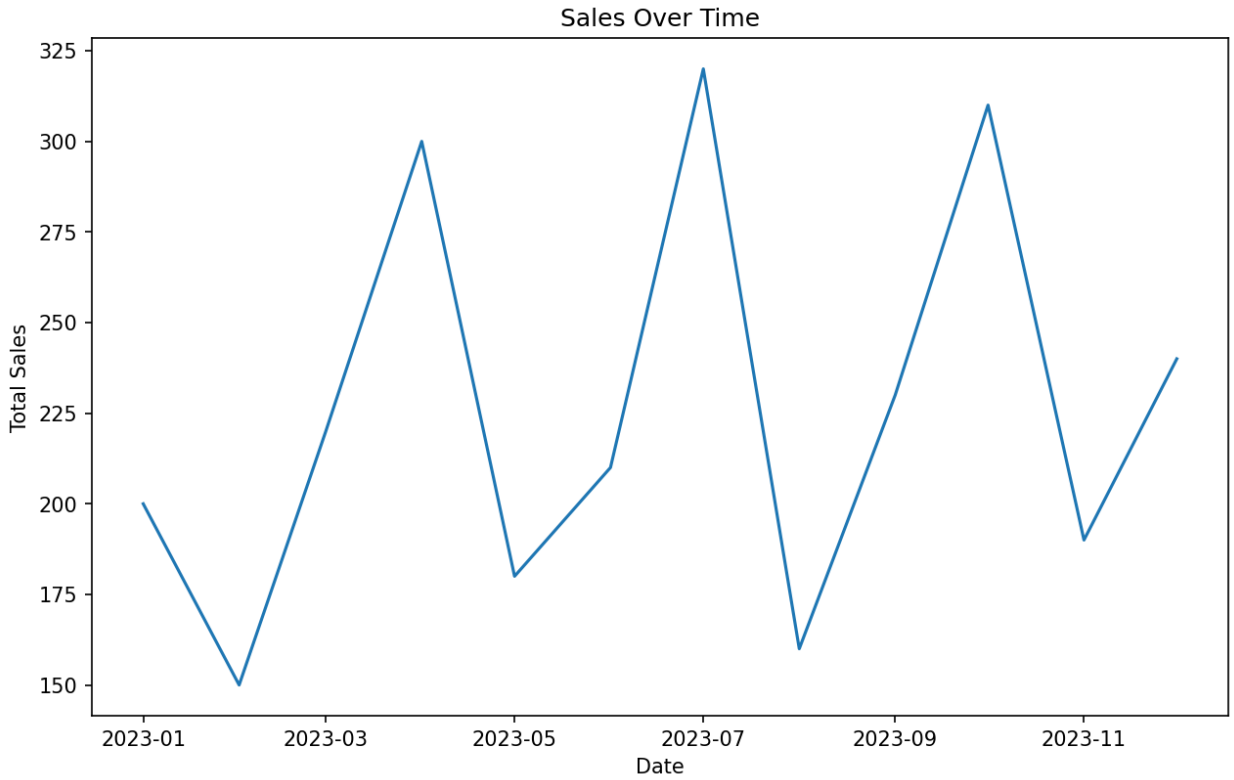
```
/tmp/ipython-input-2-cell.py:17: ChainedAssignmentError: A value is being set on a copy of a
DataFrame or Series through chained assignment using an inplace method.
Such inplace method never works to update the original DataFrame or Series, because the
intermediate object on which we are setting values always behaves as a copy (due to Copy-on-Write).

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},
inplace=True)' instead, to perform the operation inplace on the original object, or try to avoid an
 inplace operation using 'df[col] = df[col].method(value)'.

See the documentation for a more detailed explanation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/copy_on_write.html
```

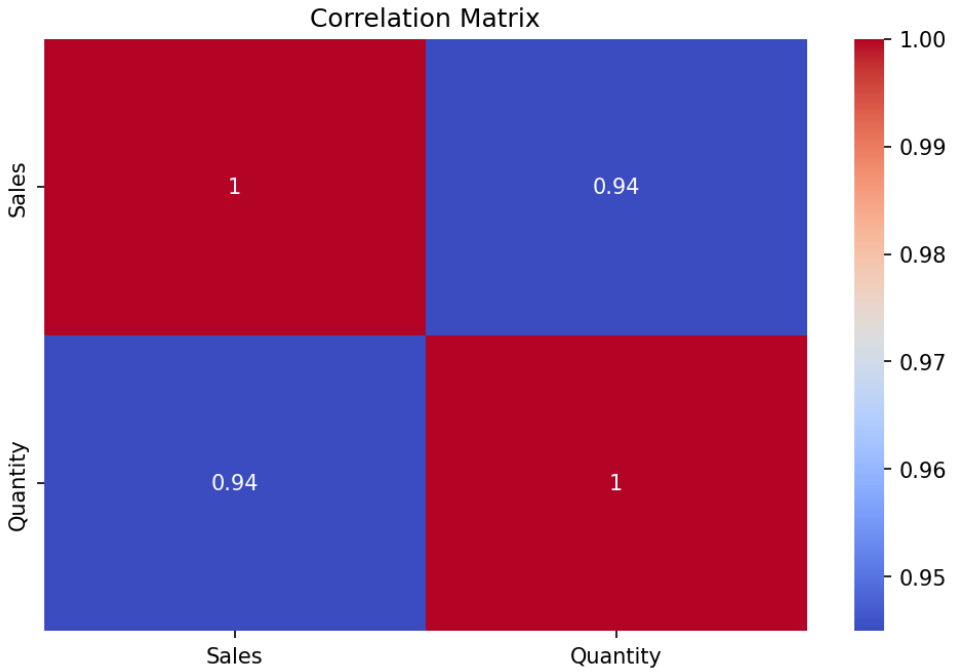
```
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95